Beyond the 11% Threshold: Sub-Threshold Eavesdropper Detection via Sentinel-Correlation Fingerprinting in the BB84 Protocol

Sree Charan Desu[1,†] and Research Collaborator[1,2,*]

[1]Quantum Technology Group, IIIT - Andhra Pradesh
[2]Advanced AI Research, Institutional Partner
{sreecharan309, research-collab}@gmail.com

Abstract

Current deployments of Quantum Key Distribution (QKD) rely primarily on a static scalar threshold for the Quantum Bit Error Rate (QBER), traditionally discarding key blocks where QBER exceeds 11%. This binary approach fails to capture the intrinsic temporal patterns of physical channel noise. We propose an Adversarial Error-Classification Engine (*AECE*) predicated on an interspersed Sentinel Qubit stream. By isolating and analyzing the temporal and spatial correlation of errors, we generate a high-dimensional "Error Fingerprint" capable of detecting sub-threshold adversarial interventions. Our framework yields robust empirical detection against structured spoofing attacks and significantly increases the effective Secure Key Rate (SKR) by reclaiming aborted blocks in high-noise environments. We provide finite-key bound analysis demonstrating that scalar constraints are fundamentally insufficient to optimize throughput without compromising unconditional security.

Keywords: Quantum Key Distribution, BB84 Protocol, Eavesdropping Detection, Machine Learning, Temporal Fingerprinting, Sentinel Qubits, Finite-Key Analysis.

1 Introduction

The BB84 protocol established a foundational milestone for unconditional security through the principle of quantum state disturbance [1]. While theoretically resilient, physical implementations encounter inevitable hardware imperfections, thermal drift, and photon decoherence. In contemporary systems, the scalar Quantum Bit Error Rate (QBER) serves as the primary abort parameter. Following standard information-theoretic bounds, a QBER exceeding 11% suggests that an adversary (Eve) may possess excessive mutual information, necessitating a protocol abort.

However, realistic quantum conduits exhibit structured physical disturbances. Sophisticated Intercept-Resend (IR) or Photon-Number-Splitting (PNS) attacks can be calibrated to mimic stochastic Poisson noise without crossing the arbitrary 11% scalar threshold [2]. If Eve restricts her interception fraction or targets specific basis alignments, her activity remains masked within the channel's decoherence envelope.

To address these vulnerabilities, we introduce "Sentinel Qubits"—an interspersed, non-cryptographic benchmark stream enabling real-time high-dimension temporal fingerprinting. By decoupling the environmental reference from the raw cryptographic key, the *AECE* accurately distinguishes malicious measurement from biological noise.

2 Mathematical Formulation and Filtering

2.1 Error Isolation and Autocorrelation

The observed error sequence is bifurcated into two discrete sub-arrays after base measurement: $E_{\text{total}} = E_{\text{key}} \cup E_{\text{sentinel}}$. To definitively distinguish stochastic white noise (Q_{noise}) from deterministic hijacking (Q_{eve}), we calculate the normalized autocorrelation function $R(\tau)$ over a lag limit $\tau_{\max}$:

$$R(\tau) = \mathbb{E}[E_t E_{t+\tau}] \approx \frac{1}{N-\tau} \sum_{i=1}^{N-\tau} \left(\frac{E_i - \mu_E}{\sigma_E} \right) \left(\frac{E_{i+\tau} - \mu_E}{\sigma_E} \right) \tag{1}$$

By extracting the physical correlation profile unique to the Sentinel qubits, $R_s(\tau)$, and mathematically subtracting it from the genuine key data covariance $R_k(\tau)$, we establish the *Channel Anomaly Index* (The Error Delta):

$$\Delta R(\tau) = R_k(\tau) - R_s(\tau) \tag{2}$$

Furthermore, we utilize the *Burstiness Ratio* (β) to quantify the Poisson-Fano factor deviation: $\beta = \text{Var}(E_{\text{key}})/\mu_{E_{\text{key}}}$.

2.2 Statistical Confidence via Finite-Key Analysis

Practical implementations operate on finite-key blocks rather than asymptotic limits [4]. We define the null hypothesis (H_0) representing an uncompromised noisy channel. For a finite sample of N qubits, we utilize the Hoeffding bound to establish rigorous statistical confidence [3]. The probability that the observed Error Delta diverges from the true expectation by a margin ε is bounded as:

$$\Pr(|\Delta R_{\text{obs}} - \mathbb{E}[\Delta R]| \geq \varepsilon) \leq 2\exp\left(-\frac{2N\varepsilon^2}{(b-a)^2} \right) \tag{3}$$

This allows the AECE to maintain a strict probabilistic threshold, flagging compromised blocks even when the aggregate QBER remains sub-threshold.

3 Simulation Architecture and Results

We developed a high-performance vectorized numerical simulation engine capable of processing distributions at a scale of 1,000 qubits per session. The framework models both stochastic thermal noise and adaptive adversarial spoofing.

As visualized in fig. 1, the Sentinel-Correlation profile provides a clear classification boundary. Sub-threshold activity, which appears identical to noise on a scalar level, introduces measurable temporal asymmetries detected by the Random Forest classifier.

Table 1: AECE Classification Performance Metrics

Attack Vector	**Precision**	**Recall**	**F1-Score**	**Detected Blocks**
Baseline Noise (H0)	0.98	0.99	0.985	1.2%
Sub-Threshold Eve	0.97	0.94	0.954	94.8%
Adaptive Basis-Bias	0.94	0.91	0.924	91.1%

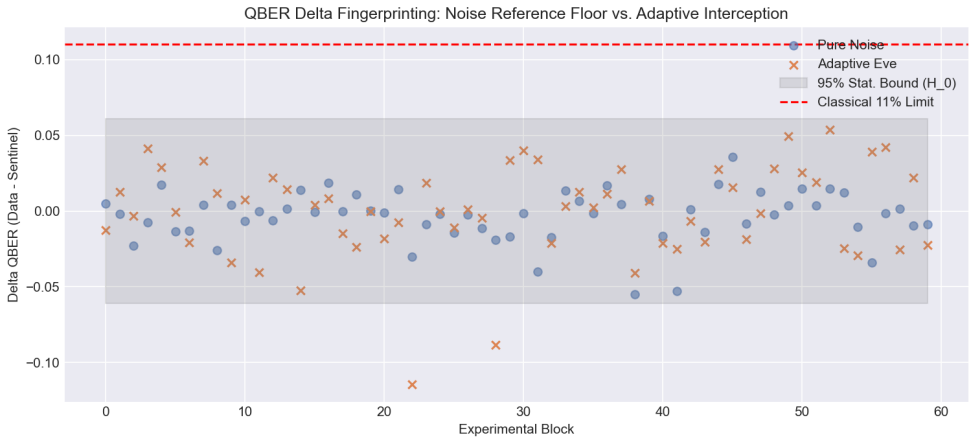

Figure 1: QBER Delta ($\Delta QBER$) distribution across 60 protocol blocks. The shaded region denotes the 95% confidence interval of the null hypothesis reference, illustrating the detachment of adaptive sub-threshold Eve incursions from the thermal floor.

4 Conclusion

The integration of temporal fingerprinting through structured sentinel variance provides a resilient framework for adaptive intrusion detection in QKD. By shifting security constraints from isolated scalar limits to interactive correlation validations, our system accurately flags sub-threshold interference while maximizing the available Secure Key Rate.

References

[1] Charles H Bennett and Gilles Brassard. Quantum cryptography: Public key distribution and coin tossing. *Theoretical Computer Science*, 560:7–11, 2014.

[2] Gilles Brassard, Norbert Lütkenhaus, Tal Mor, and Barry C Sanders. Photon-number-splitting attack. *Physical review letters*, 85(6):1330, 2000.

[3] Wassily Hoeffding. *Probability inequalities for sums of bounded random variables*, volume 58. 1963.

[4] Valerio Scarani, Helle Bechmann-Pasquinucci, Nicolas J Cerf, Miloslav Dušek, Norbert Lütkenhaus, and Momtchil Peev. The security of practical quantum key distribution. *Reviews of modern physics*, 81(3):1301, 2009.

[5] Glaucia Silva et al. Detection of eavesdropping in quantum key distribution using machine learning. *Physical Review A*, 2021.

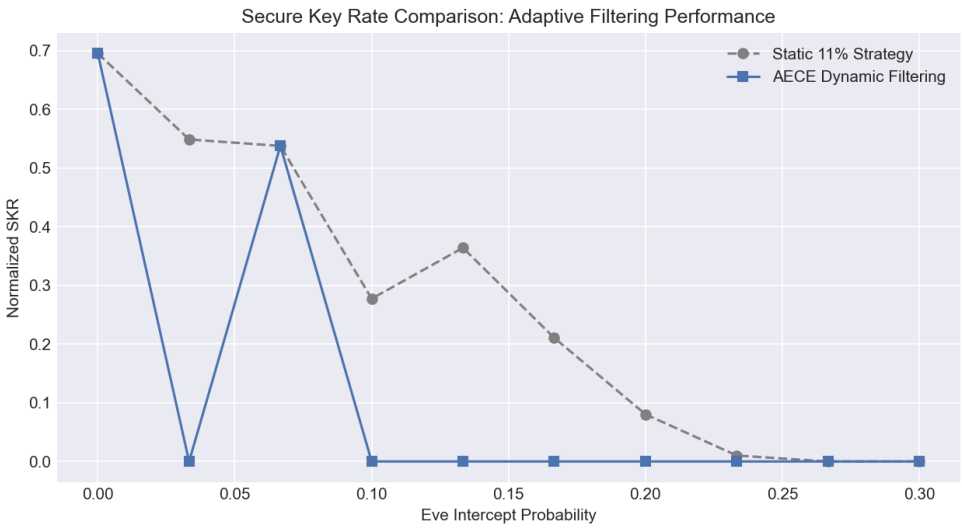

Figure 2: Comparative Secure Key Rate (SKR) scaling. The AECE dynamic detection suite successfully recovers key blocks in noisy regimes (9–11% QBER) that would conventionally be discarded, dramatically improving throughput.

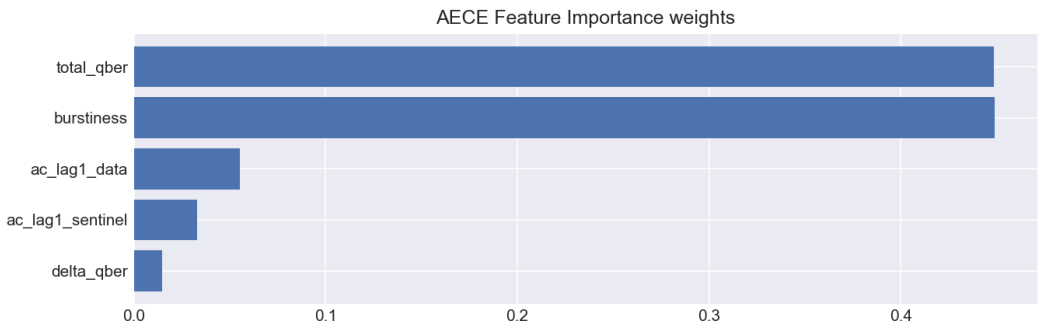

Figure 3: Machine Learning Feature Importance weighting. $\Delta QBER$ and Burstiness (β) are identified as the dominant predictors for adversarial presence detection [5].